Explainable Accounting-Fraud Detection

Extending Bao et al. (Journal of Accounting Research, 2020) with scale-free features, an ensemble and a decision step

Final Year Project · Midterm evaluation · 2026

From raw filings to an explained fraud score



Data

146k firm-years, 1991–2014, SEC fraud labels

Features

133 scale-free ratios and within-year ranks

Models

LightGBM, XGBoost and logistic regression

Evaluation

Train on the past, test on a later year

Explain

SHAP: why each firm was flagged

Deploy

Calibrated probability and a scoring script

The feature pool is fixed, not re-selected each year. This is a tested design choice, not a shortcut: the next slide gives the argument and the evidence.

Why the features are fixed, not dynamic

1

Chosen once, on validation years

Picked as a set on 1999–2001 folds. The test years never influenced the choice.

2

Dynamic options were tested fairly

12 selection and generation methods ran inside every training window, under rules fixed before any run. None cleared the bar.

3

Fixed means stable and explainable

Every year's model uses the same features, so SHAP explanations can be compared and audited across years.

Direct test on the test years

+0.0008

AUC change from dynamic selection. 95% CI −0.004 to +0.006: no measurable difference.

59–200

features kept for the same year, depending only on the random seed.

Dynamic selection adds cost and instability without a measurable gain.

Bao et al. (2020): machine learning finds fraud

Bao, Ke, Li, Yu & Zhang
Journal of Accounting Research, 58(1), 2020

Goal
Rank U.S. public firms by their risk of SEC fraud enforcement (AAERs), one year ahead

Method
RUSBoost: boosted trees with random under-sampling, built for rare events

Finding
Raw items beat expert ratios, and the model beats earlier benchmarks

28
raw financial items as inputs

0.66%
of firm-years are fraud

≈0.72
AUC on 2003–2008 test years

2 yrs
gap between training and test

How the paper tests: train on the past, predict the future

Expanding window, one row per test year



Expanding window

Retrain for every test year on all earlier data

2-year gap

Recent fraud is not public yet, so year T−1 is left out

Serial-fraud masking

A case running into the test year is removed from training

We rebuilt the paper in Python and matched it

Data	146,045 firm-years from 18,444 firms, 1991–2014
Fraud labels	964 fraud firm-years in 412 SEC cases
Model	RUSBoost: 300 trees, leaf size 5, learning rate 0.1
Sampling	Non-fraud under-sampled to 1:1 in every round
Tuning	Train 1991–1999, validate on 2001

Paper reports

≈0.72 AUC

Our rebuild

0.705–0.724

Reproduced: a fair baseline for every later gain

Baseline used from here on: 0.705, the paper's own tree settings.

Where the field stood before this project

2010 ≈0.63 AUC

Cecchini et al.
Support vector machine with a financial kernel

2011 ≈0.69 AUC

Dechow et al.
F-score: logistic model on 14 expert ratios

2017 used here

Lundberg & Lee
SHAP: consistent, per-prediction explanations

2020 ≈0.72 AUC

Bao et al.
RUSBoost on 28 raw items beats both benchmarks

2021 future work

Bertomeu et al.
Audit and governance data add signal

2023 tested here

Zhang et al. (OpenFE)
Automatic feature generation

Five gaps in the paper, five fixes

Scale bias	**Black box**	**No decision step**	**Thin statistics**	**Not deployable**
Raw dollar items let firm size drive risk	No reason given for a flag	A ranking, with no cutoff	No confidence intervals	No way to score new firms
Scale-free ratios ranked within each year	**SHAP explanation for each flagged firm**	**Budgets, lift and calibrated probabilities**	**Bootstrap clustered by fraud case**	**Deployable model and scoring script**

Features that measure risk, not firm size

Removed

The 28 raw dollar items, which mostly measure size

Added

Beneish M-score parts, Altman Z, accrual, growth and financing-need ratios

Ranked

Each ratio ranked within its own year, so 1991 and 2014 are comparable

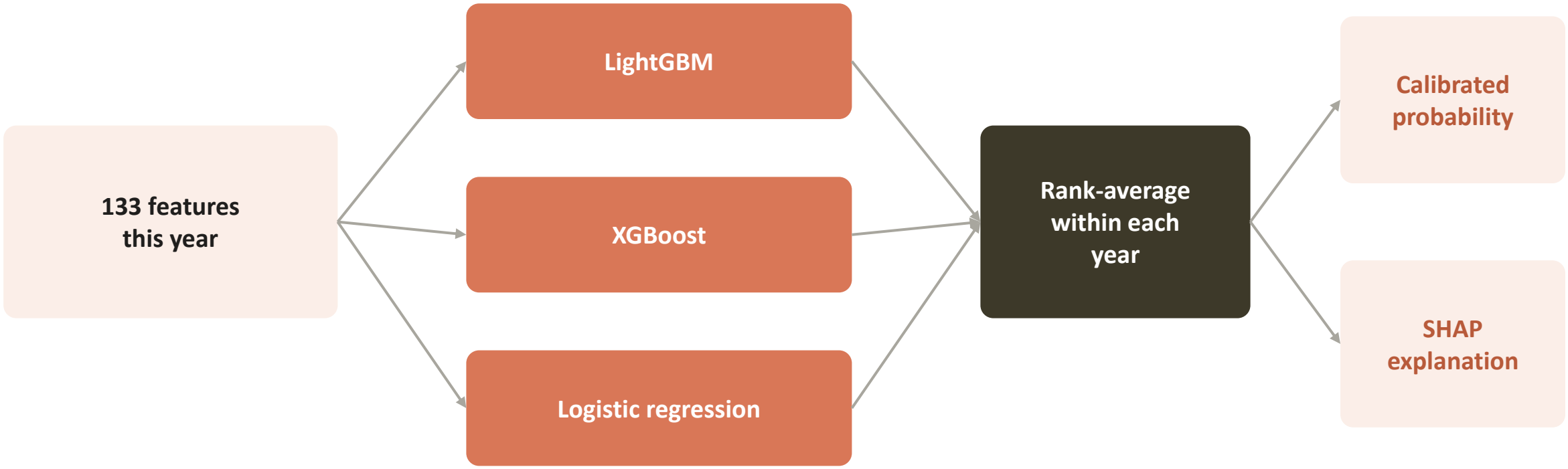
Chosen on validation years

Raw 28 items: 0.701 validation AUC. Our 133 features: 0.745

No peeking at the test years

Picked on 1999–2001 folds; the test years were never used

Three models, one ranked and explained score



Tuned only on data before 2003; the paper's model is never a member.

Headline results, test years 2003–2012

0.705 → 0.735

ROC AUC
Paper baseline to our ensemble

9.8% → 15.2%

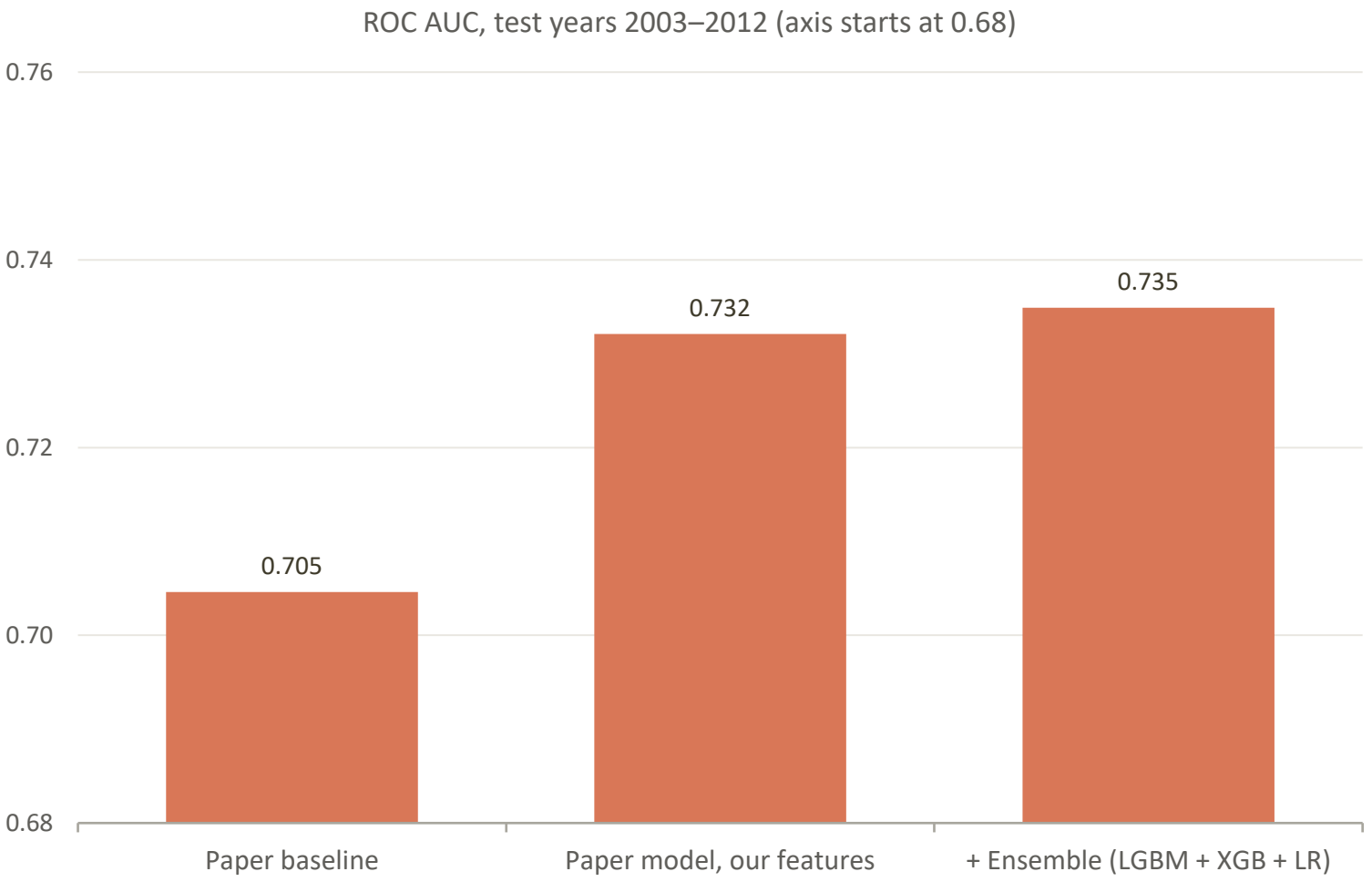
Frauds caught in the top 5%
About 55% more than the baseline

1.9 per 100

Frauds found per 100 reviews
At a 1% budget; baseline 1.8, random 0.6

The features alone give a significant AUC gain: +0.0275, 95% CI +0.010 to +0.046 (bootstrap clustered by fraud case).

Where the gain comes from



+0.0275

from the features alone

+0.003

from the ensemble, within noise

Most of the gain comes from better inputs, not a fancier model.

What the results do not show

A ranking tool, not a detector

About 98 of every 100 flagged firms are not labelled fraud. It is a way to prioritise investigations.

Firm size still matters

Sales and assets rank highest in SHAP, even without raw dollar items.

Why not 0.8?

Weaker test rules add about 0.09 AUC on the same models. We keep the strict rules.

Labels are incomplete

Only SEC-caught fraud is labelled, and recent years keep gaining labels: 69 frauds in 2003, 4 in 2014.

Remaining work

NEXT STEP	HOW
Size-neutral model	Drop size features or model within size buckets; check that precision at 1% holds
Cost-based cutoff	Set the review budget from the cost of a review against the loss avoided per fraud caught
Stronger statistics	Per-year significance tests and calibration checks on the stored predictions
Audit and governance data	Needs paid WRDS data: to be requested through the university; if unavailable, documented as the main limit
Final report and viva	Write-up, result tables and a live demo with the scoring script

Thank you

Questions are welcome

- Reproduced the JAR 2020 baseline
- Raised ROC AUC from 0.705 to 0.735 with scale-free features and an ensemble
- Kept the features fixed after testing 12 dynamic alternatives